

# Phy101 Final Term Short Notes By Pin2.

## Electric potential energy

Electric potential energy and electric potential are closely related concepts. The electric potential turns out to be just the electric potential energy per unit charge. This relationship is similar to that between electric force and the electric field, which is the electric force per unit charge.

Electric potential energy is a scalar quantity.

### APPLICATION

#### Automobile Batteries

An application of potential difference is the 12-V battery found in an automobile. Such a battery maintains a potential difference across its terminals, with the positive terminal 12 V higher in potential than the negative terminal. In practice the negative terminal is usually connected to the metal body of the car, which can be considered to be at a potential of zero volts. The battery provides the electrical current necessary to operate headlights, a radio, power windows, motors, and so forth.

## CAPACITANCE

A **capacitor** is a device used in a variety of electric circuits, such as to tune the frequency of radio receivers, eliminate sparking in automobile ignition systems, or store short-term energy for rapid release in electronic flash units.

It consists of two parallel metal plates separated by a distance  $d$ . Used in an electric circuit, the plates are connected to the positive and negative terminals of a battery or some other voltage source. When this connection is made, electrons are pulled off one of the plates, leaving it with a charge of  $-Q$ , and are transferred through the battery to the other plate, leaving it with a charge of  $+Q$ , as shown in the figure. The transfer of charge stops when the potential difference across the plates equals the potential difference of the battery. A charged capacitor is a device that stores energy that can be reclaimed when needed for a specific application.

The capacitance  $C$  of a capacitor is the ratio of the magnitude of the charge on either conductor (plate) to the magnitude of the potential difference between the conductors (plates):

## APPLICATIONS

Camera Flash Attachments

Computer Keyboards

Electrostatic Confinement

## COMBINATIONS OF CAPACITORS

Two or more capacitors can be combined in circuits in several ways, but most reduce to two simple configurations, called *parallel* and *series*.

Two **Capacitors** are said to be in *parallel*.

The left plate of each capacitor is connected to the positive terminal of the battery by a conducting wire, so the left plates are at the same potential. In the same way, the right plates, both connected to the negative terminal of the battery, are also at the same potential. This means that **capacitors in parallel both have the same potential difference  $V$  across them.**

## Capacitors in Series

two capacitors connected in *series*,

**For a series combination of capacitors, the magnitude of the charge must be the same on all the plates.** When a battery is connected to the circuit, electrons with total charge  $Q$  are transferred from the left plate of  $C_1$  to the right plate of  $C_2$  through the battery, leaving the left plate of  $C_1$  with a charge of  $Q$ . As a consequence, the magnitudes of the charges on the left plate of  $C_1$  and the right plate of  $C_2$  must be the same. Now consider the right plate of  $C_1$  and the left plate of  $C_2$ , in the middle. These plates are not connected to the battery (because of the gap across the plates) and, taken together, are electrically neutral. The charge of  $Q$  on the left plate of  $C_1$ , however, attracts negative charges to the right plate of  $C_1$ . These charges will continue to accumulate until the left and right plates of  $C_1$ , taken together, become electrically neutral, which means that the charge on the right plate of  $C_1$  is  $Q$ . This negative charge could only have come from the left plate of  $C_2$ , so  $C_2$  has a charge of  $Q$ .

## **DIELECTRICS**

**A dielectric is an insulating material, such as rubber, plastic, or waxed paper.**

When a dielectric is inserted between the plates of a capacitor, the capacitance increases. If the dielectric completely fills the space between the plates, the capacitance is multiplied by the factor  $k$ , called the **dielectric constant**.

## **ELECTRIC CURRENT**

**The current is the rate at which charge flows through this surface.**

**SI unit: coulomb/second (C/s), or the ampere (A)**

**The instantaneous current  $I$  is the limit of the average current as the time interval goes to zero.**

**SI unit: coulomb/second (C/s), or the ampere (A)**

## **RESISTANCE, RESISTIVITY, AND OHM'S LAW**

When a voltage (potential difference)  $V$  is applied across the ends of a metallic conductor, the current in the conductor is found to be proportional to the applied voltage;  $I \propto V$ . If the proportionality holds, we can write  $V = IR$ , where the proportionality constant  $R$  is called the **resistance** of the conductor. In fact, we define the **resistance** as the ratio of the voltage across the conductor to the current it carries.

$$R = \frac{\Delta V}{I}$$

**Resistance has SI units of volts per ampere, called ohms  $\Omega$ .** If a potential difference of 1 V across a conductor produces a current of 1 A, the resistance of the conductor

The concepts of electric current, voltage, and resistance can be compared to the flow of water in a river. As water flows downhill in a river of constant width and depth, the flow rate (water current) depends on the steepness of descent of the river and the effects of rocks, the riverbank, and other obstructions.

**The resistance remains constant over a wide range of applied voltages or currents.**

This statement is known as **Ohm's law**, after Georg Simon Ohm (1789–1854), who was the first to conduct a systematic study of electrical resistance.

Ohm's law is given by

$$\Delta V = IR$$

## **SUPERCONDUCTORS**

There is a class of metals and compounds with resistances that fall virtually to zero below a certain temperature  $T_c$  called the *critical temperature*. These materials are known as **superconductor**.

## **ELECTRICAL ACTIVITY IN THE HEART**

The voltage pulses can be recorded on an instrument called an **electrocardiograph**, and the pattern recorded by this instrument is called an **electrocardiogram** (EKG).

## **SOURCES OF EMF**

A current is maintained in a closed circuit by a source of emf. Among such sources are any devices (for example, batteries and generators) that increase the potential energy of the circulating charges. A source of emf can be thought of as a "charge pump" that forces electrons to move in a direction opposite the electrostatic field inside the source. The emf  $\mathcal{E}$  of a source is the work done per unit charge; hence the SI unit of emf is the volt.

## **RESISTORS IN SERIES**

When two or more resistors are connected end to end,

**The current is the same in the two resistors because any charge that flows through  $R_1$  must also flow through  $R_2$ .**

## **Kirchhoff's rules**

1. The sum of the currents entering any junction must equal the sum of the currents leaving that junction. (This rule is often referred to as the **junction rule**.)
2. The sum of the potential differences across all the elements around any closed circuit loop must be zero. (This rule is usually called the **loop rule**.)

## **MAGNETS**

Most people have had experience with some form of magnet. You are most likely familiar with the common iron horseshoe magnet that can pick up iron-containing objects such as paper clips and nails. In the discussion that follows, we assume the magnet has the shape of a bar. Iron objects are most strongly attracted to either end of such a bar magnet, called its **poles**. One end is called the **north pole** and the other the **south pole**.

**Like poles repel each other and unlike poles attract each other.**

**Soft magnetic materials**, such as iron, are easily magnetized, but also tend to lose their magnetism easily. In contrast, **hard magnetic materials**, such as cobalt and nickel, are difficult to magnetize, but tend to retain their magnetism.

## **EARTH'S MAGNETIC FIELD**

A small bar magnet is said to have north and south poles, but it's more accurate to say it has a "north-seeking" pole and a "south-seeking" pole. By these expressions, we mean that if such a magnet is used as a compass, one end will "seek," or point to, the geographic North Pole of Earth and the other end will "seek," or point to, the geographic South Pole of Earth. We conclude that **the geographic North Pole of Earth corresponds to a magnetic south pole, and the geographic South Pole of Earth corresponds to a magnetic north pole**.



## MAGNETIC FIELDS

Experiments show that a stationary charged particle doesn't interact with a static magnetic field. When a charged particle is moving through a magnetic field, however,

a magnetic force acts on it. This force has its maximum value when the charge moves in a direction perpendicular to the magnetic field lines, decreases in value at other angles, and becomes zero when the particle moves along the field lines.

This is quite different from the electric force, which exerts a force on a charged particle whether it's moving or at rest. Further, the electric force is directed parallel to the electric field whereas the magnetic force on a moving charge is directed perpendicular to the magnetic field.

SI unit of magnetic field is the tesla (T), also called the weber (Wb) per square meter ( $1 \text{ T} = 1 \text{ Wb/m}^2$ ).

## MAGNETIC FORCE ON A CURRENT-CARRYING CONDUCTOR

If a magnetic field exerts a force on a single charged particle when it moves through a magnetic field, it should be no surprise that magnetic forces are exerted on a current-carrying wire as well. Because the current is a collection

of many charged particles in motion, the resultant force on the wire is due to the sum of the individual forces on the charged particles. The force on the particles is transmitted to the "bulk" of the wire through collisions with the atoms making up the wire.

**Definition of the ampere** : If two long, parallel wires 1 m apart carry the same current and the magnetic force per unit length on each wire is  $2 \times 10^{-7} \text{ N/m}$ , the current is defined to be 1 A.

**Definition of the coulomb** : If a conductor carries a steady current of 1 A, the quantity of charge that flows through any cross section in 1 s is 1 C.

### Magnetic Field of a Solenoid

If a long, straight wire is bent into a coil of several closely spaced loops, the resulting device is a solenoid, often called an electromagnet.

### Types of Magnetic Materials

**Paramagnetic** materials also have magnetic moments that tend to align with an externally applied magnetic field, but the response is extremely weak compared with that of ferromagnetic materials. Examples of paramagnetic substances are aluminum, calcium, and platinum. A ferromagnetic material can become paramagnetic when warmed to a certain critical temperature, the Curie temperature, that depends on the material.

In **diamagnetic** materials, an externally applied magnetic field induces a very weak magnetization that is opposite the applied field. Ordinarily diamagnetism isn't observed because paramagnetic and ferromagnetic effects are far stronger. In Figure 19.39, however, a very high magnetic field exerts a levitating force on the diamagnetic water molecules in a frog.

### Magnetic Force on a Current-Carrying

#### Conductor

If a straight conductor of length  $\ell$  carries current  $I$ , the magnetic force on that conductor when it is placed in a uniform external magnetic field  $\mathbf{B}$  is

$$F = B\ell \sin \phi$$

### Magnetic Flux

The magnetic flux  $\phi_B$  through a loop of wire with area  $A$  is defined by

$$\phi_B = BA \cos \theta$$

The value of the magnetic flux is proportional to the total number of lines passing through the loop.

### **Lenz's law:**

The current caused by the induced emf travels in the direction that creates a magnetic field with flux opposing the change in the original flux through the circuit.

### **GENERATORS**

Generators and motors are important practical devices that operate on the principle of electromagnetic induction. First, consider the **alternating-current (AC) generator**, a device that converts mechanical energy to electrical energy. In its simplest form, the AC generator consists of a wire loop rotated in a magnetic field by some external means.

#### **Motors and Back emf**

Motors are devices that convert electrical energy to mechanical energy. Essentially, **a motor is a generator run in reverse**: instead of a current being generated by a rotating loop, a current is supplied to the loop by a source of emf, and the magnetic torque on the current-carrying loop causes it to rotate.

### **MAXWELL'S PREDICTIONS**

1. Electric field lines originate on positive charges and terminate on negative charges.
2. Magnetic field lines always form closed loops; they don't begin or end anywhere.
3. A varying magnetic field induces an emf and hence an electric field. This fact is a statement of Faraday's law (Chapter 20).
4. Magnetic fields are generated by moving charges (or currents), as summarized in Ampère's law.

### **Waves**

**Radio waves**, are the result of charges accelerating through conducting wires. They are, of course, used in radio and television communication systems.

**Microwaves** (short-wavelength radio waves) have wavelengths ranging between about 1 mm and 30 cm and are generated by electronic devices.

**Infrared waves** (sometimes incorrectly called "heat waves"), produced by hot objects and molecules, have wavelengths ranging from about 1 mm to the longest wavelength of visible light.

**Visible light**, the most familiar form of electromagnetic waves, may be defined as the part of the spectrum that is detected by the human eye. Light is produced by the rearrangement of electrons in atoms and molecules.

**Gamma rays**—electromagnetic waves emitted by radioactive nuclei—have wavelengths ranging from about  $10^{-10}$  m to less than  $10^{-14}$  m. They are highly penetrating and cause serious damage when absorbed by living tissues.

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Best Of Luck For the Exams.